

DTF via Robust Scaling: Replication Code for Simulation 2

Blinded

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Initialization

```
library(robustDIF)
library(dplyr)
library(tidyr)
library(ggplot2)
library(see)
library(mirt)

knitr::opts_chunk$set(fig.width = 11.5, fig.height = 5.4)

check_archive_robustDIF <- function(version = "0.2.0.20260708",
                                     archive.name = "review-archive") {
  installed_version <- as.character(utils::packageVersion("robustDIF"))
  if (installed_version != version) {
    stop(
      "This report expects robustDIF ", version,
      ", but found robustDIF ", installed_version,
      ". Install robustDIF from the archive branch.",
      call. = FALSE)
  }

  desc <- utils::packageDescription("robustDIF")
  installed_archive <- desc[["Config/robustDIF/archive-name"]]
  if (!identical(installed_archive, archive.name)) {
    stop(
      "This report expects robustDIF archive '", archive.name,
```

```

    "', but found archive label '", installed_archive,
    "'. Install robustDIF from the archive branch.",
    call. = FALSE)
}
invisible(installed_version)
}

check_archive_robustDIF()

find_results_dir <- function(candidates, pattern, min_files, label) {
  candidates <- unique(candidates)
  n_files <- vapply(
    candidates,
    function(path) {
      if (!dir.exists(path)) return(0L)
      length(list.files(path, pattern = pattern, full.names = TRUE))
    },
    integer(1))

  if (!any(n_files >= min_files)) {
    stop(
      "Could not find the ", label, " result files. Expected at least ",
      min_files, " files matching ", pattern, " in one of: ",
      paste(candidates, collapse = ", "),
      call. = FALSE)
  }

  normalizePath(candidates[which(n_files >= min_files)[1]], mustWork = TRUE)
}

sim2_pattern <- "^sim2_target_A.*_Di.*_ndif[0-9]+[.]rds$"
results_dir <- find_results_dir(
  c(
    "../results/sim2_target_delta",
    "results/sim2_target_delta",
    "../..paper/results/sim2_target_delta",
    "../...paper/results/sim2_target_delta",
    "paper/results/sim2_target_delta"),
  pattern = sim2_pattern,
  min_files = 66L,
  label = "Simulation 2")

```

```

archive_root_candidates <- c("../", "replication")
archive_root <- archive_root_candidates[
  dir.exists(file.path(archive_root_candidates, "qmd")) &
  file.exists(file.path(archive_root_candidates, "README.md"))][1]
if (is.na(archive_root)) archive_root <- ".."

figure_dir <- file.path(archive_root, "figures/sim2_target_delta")
dir.create(figure_dir, recursive = TRUE, showWarnings = FALSE)

method_levels <- c("RDIF", "True")

method_cols <- c(
  "RDIF" = "#7BAFD4",
  "True" = "grey50")

alpha <- 0.05
ndif_levels <- seq(0, 20, by = 2)
p_dif_breaks <- seq(0, 1, by = 0.2)
target_levels <- c(0.25, 0.5)
slope_levels <- c(0.5, 1, 1.5)

mean_or_na <- function(x) {
  if (all(is.na(x))) NA_real_ else mean(x, na.rm = TRUE)
}

sd_or_na <- function(x) {
  if (sum(!is.na(x)) < 2) NA_real_ else sd(x, na.rm = TRUE)
}

breakdown_band <- function(data) {
  data |>
    distinct(target_Delta_i_public, slope_cell_public, target_Delta_i_num) |>
    mutate(
      xmin = 0.5 * n_items,
      xmax = Inf,
      ymin = -Inf,
      ymax = Inf)
}

ndif_x_scale <- function(data) {
  item_effect <- unique(data$target_Delta_i_num)
  if (length(item_effect) != 1L || is.na(item_effect)) {

```

```

    stop("Expected a single item-level effect size.", call. = FALSE)
  }

n_dif_breaks <- seq(0, n_items, by = 4)
true_delta_breaks <- n_dif_breaks * item_effect / n_items

scale_x_continuous(
  breaks = n_dif_breaks,
  labels = n_dif_breaks,
  expand = expansion(add = c(0.8, 0.8)),
  sec.axis = sec_axis(
    ~ . * item_effect / n_items,
    breaks = true_delta_breaks,
    labels = sprintf("%.2f", true_delta_breaks),
    name = "True Delta")
)

sim2_public_facets <- function(free_y = FALSE) {
  facet_wrap(
    vars(slope_cell_public),
    ncol = 3,
    scales = if (free_y) "free" else "free_x",
    axes = "all_x",
    axis.labels = "all_x")
}

plot_by_effect <- function(data, plot_fun, file_prefix = NULL,
                           width = 10, height = 4.8, dpi = 200) {
  effects <- levels(droplevels(data$target_Delta_i_public))
  plots <- vector("list", length(effects))
  names(plots) <- effects

  for (effect in effects) {
    plot_data <- data |>
      filter(target_Delta_i_public == effect) |>
      droplevels()

    plot <- plot_fun(plot_data) +
      labs(title = effect)

    print(plot)
    plots[[effect]] <- plot
  }
}

```

```

    if (!is.null(file_prefix)) {
      file_effect <- gsub("[^A-Za-z0-9]+", "_", effect)
      file_effect <- gsub("^|_$", "", file_effect)
      ggsave(
        file.path(ffigure_dir, paste0(file_prefix, "_", file_effect, ".png")),
        plot,
        width = width,
        height = height,
        dpi = dpi)
    }
  }

  invisible(plots)
}

public_plot_theme <- theme(
  panel.grid.minor = element_blank(),
  panel.spacing.x = grid::unit(1.2, "lines"),
  plot.margin = margin(t = 12, r = 8, b = 8, l = 8))

theme_set(theme_bw(base_size = 14))

```

Run Sims

The production run was carried out condition-by-condition. Each condition was saved separately as an .rds file under `results/sim2_target_delta`, so the simulation can be restarted without overwriting completed conditions.

```

SIM2_TARGET_REPS=500 SIM2_TARGET_CORES=10 SIM2_TARGET_SEED=20260724 \
SIM2_TARGET_OVERWRITE=TRUE SIM2_TARGET_RUN_LABEL=production \
Rscript run_sim2.R

```

The initial production run used `n.dif = 5` to include exactly 25 percent DIF items. We subsequently ran the missing `n.dif = 6` cells and use the even-count grid below for reporting.

```

SIM2_TARGET_REPS=500 SIM2_TARGET_CORES=10 SIM2_TARGET_SEED=20260724 \
SIM2_TARGET_OVERWRITE=FALSE SIM2_TARGET_RUN_LABEL=add_ndif6 \
Rscript run_sim2.R

```

Simulation 2 uses 500 replications in each reported cell of a 3 x 2 x 11 grid:

- slope multiplier for true DIF items: 0.5, 1, 1.5;
- target item-level effect size for true DIF items: `Delta_i` = 0.25, 0.50;
- number of DIF items: 0, 2, 4, ..., 20;
- total number of items: 20;
- sample size per group: 500;
- reference latent mean and variance fixed at 0 and 1;
- comparison latent mean sampled from $U(-0.25, 0.25)$;
- comparison latent variance sampled from $U(0.5, 1.5)$.

For `n.dif > 0`, the group-2 slope is multiplied by the slope factor and the group-2 intercept is solved item-by-item so the true DIF items have the requested `Delta_i` on the `d_fun3` scale. For `n.dif = 0`, there are no DIF items; these cells provide the Type I error baseline.

The reported methods are `RDIF` and `True`, shown in alphabetical order. Here `RDIF` denotes the BH-start version of RDIF, and `True` denotes the oracle unit-weighted procedure using the known DIF pattern. The LRT and post-RDIF unit-weight procedures are omitted from this larger grid.

Load Results

```
sim_files <- list.files(
  results_dir,
  pattern = sim2_pattern,
  full.names = TRUE)

sim_list <- lapply(sim_files, readRDS)

sim_delta_all <- bind_rows(lapply(sim_list, `[`, "delta")) |>
  mutate(
    method = factor(as.character(method), levels = method_levels),
    n_dif = as.integer(n_dif),
    n_items = as.integer(n_items),
    p_dif = n_dif / n_items,
    target_Delta_i_num = as.numeric(target_Delta_i),
    target_Delta_i = factor(
      target_Delta_i_num,
      levels = target_levels,
      labels = c("Delta_i = .25", "Delta_i = .50")),
    target_Delta_i_public = factor(
      target_Delta_i_num,
      levels = target_levels,
```

```

      labels = c("Item-Level Effect = 0.25", "Item-Level Effect = 0.50")),
    slope_multiplier = as.numeric(slope_multiplier),
    slope_cell = factor(
      paste0("A = ", slope_multiplier),
      levels = paste0("A = ", slope_levels)),
    slope_cell_public = factor(
      paste0("Slope Multiplier = ", slope_multiplier),
      levels = paste0("Slope Multiplier = ", slope_levels)),
    breakdown = p_dif >= 0.5)

sim_delta <- sim_delta_all |>
  filter(n_dif %in% ndif_levels)

sim_items_all <- bind_rows(lapply(sim_list, `[`, "items")) |>
  mutate(
    method = factor(as.character(method), levels = method_levels),
    n_dif = as.integer(n_dif),
    n_items = as.integer(n_items),
    p_dif = n_dif / n_items,
    target_Delta_i_num = as.numeric(target_Delta_i),
    target_Delta_i = factor(
      target_Delta_i_num,
      levels = target_levels,
      labels = c("Delta_i = .25", "Delta_i = .50")),
    target_Delta_i_public = factor(
      target_Delta_i_num,
      levels = target_levels,
      labels = c("Item-Level Effect = 0.25", "Item-Level Effect = 0.50")),
    slope_multiplier = as.numeric(slope_multiplier),
    slope_cell = factor(
      paste0("A = ", slope_multiplier),
      levels = paste0("A = ", slope_levels)),
    slope_cell_public = factor(
      paste0("Slope Multiplier = ", slope_multiplier),
      levels = paste0("Slope Multiplier = ", slope_levels)),
    breakdown = p_dif >= 0.5)

sim_items <- sim_items_all |>
  filter(n_dif %in% ndif_levels)

run_info <- bind_rows(lapply(sim_list, function(x) {
  data.frame(

```

```

    n.reps = x$run$n.reps,
    n.cores = x$run$n.cores,
    base.seed = x$run$base.seed,
    condition.seed = x$run$condition.seed,
    save.mle = x$run$save.mle,
    save.fits = x$run$save.fits,
    completed = x$run$completed)
  )))

status_files <- file.path(
  results_dir,
  c(
    "sim2_target_delta_status_production.csv",
    "sim2_target_delta_status_add_ndif6.csv"))
sim_status <- bind_rows(lapply(status_files[file.exists(status_files)], read.csv))

n_reps <- unique(sim_delta |>
  count(slope_multiplier, target_Delta_i, n_dif, method, name = "n") |>
  pull(n))
n_items <- unique(sim_delta$n_items)

stopifnot(nrow(distinct(sim_delta, slope_multiplier, target_Delta_i, n_dif)) == 66L)

```

Item-Level Diagnostics

```

item_diag <- sim_items |>
  mutate(flagged_bh = p_adjusted <= alpha) |>
  summarize(
    flagged_rate = mean(flagged_bh, na.rm = TRUE),
    mean_p = mean_or_na(p_value),
    mean_p_adjusted = mean_or_na(p_adjusted),
    mean_weight = mean_or_na(rdif_weight),
    mean_k = mean_or_na(k_i),
    true_Delta = mean_or_na(true_Delta),
    .by = c(
      slope_multiplier, slope_cell, slope_cell_public,
      target_Delta_i, target_Delta_i_num, target_Delta_i_public,
      n_dif, p_dif, true_dif)) |>
  mutate(

```



```

    diagnostic = ifelse(true_dif, "True positive rate", "False positive rate"))

knitr::kable(
  item_diag |>
    select(
      -target_Delta_i_num,
      -target_Delta_i_public,
      -slope_cell_public,
      -true_Delta),
  digits = 3)

```

slope_multiplier	slope_cell_public	target_Delta_i_public	target_Delta_i_public	slope_cell_public	true_Delta	lagged_mean	mean_p	adjusted_mean	height	diagnostic	
0.5	A = 0.5	Delta_i = .25	0	0.0	FALSE	0.004	0.505	0.809	0.797	2.298	False positive rate
0.5	A = 0.5	Delta_i = .25	2	0.1	FALSE	0.007	0.502	0.781	0.789	2.273	False positive rate
0.5	A = 0.5	Delta_i = .25	2	0.1	TRUE	0.082	0.288	0.586	0.548	2.530	True positive rate
0.5	A = 0.5	Delta_i = .25	4	0.2	FALSE	0.009	0.503	0.760	0.782	2.255	False positive rate
0.5	A = 0.5	Delta_i = .25	4	0.2	TRUE	0.079	0.306	0.583	0.566	2.472	True positive rate
0.5	A = 0.5	Delta_i = .25	6	0.3	TRUE	0.078	0.327	0.582	0.581	2.423	True positive rate
0.5	A = 0.5	Delta_i = .25	6	0.3	FALSE	0.013	0.496	0.736	0.773	2.245	False positive rate
0.5	A = 0.5	Delta_i = .25	8	0.4	TRUE	0.072	0.353	0.586	0.603	2.374	True positive rate
0.5	A = 0.5	Delta_i = .25	8	0.4	FALSE	0.021	0.485	0.708	0.756	2.248	False positive rate

slope_multiplier	cell	target_Delta_i	diff	true_diff	lagged_mean	mean_p	adjusted	height	diagnostic		
0.5	A = 0.5	Delta_i = .25	10	0.5	FALSE	0.041	0.444	0.661	0.712	2.285	False positive rate
0.5	A = 0.5	Delta_i = .25	10	0.5	TRUE	0.046	0.399	0.623	0.652	2.311	True positive rate
0.5	A = 0.5	Delta_i = .25	12	0.6	TRUE	0.035	0.446	0.663	0.697	2.265	True positive rate
0.5	A = 0.5	Delta_i = .25	12	0.6	FALSE	0.080	0.386	0.601	0.645	2.348	False positive rate
0.5	A = 0.5	Delta_i = .25	14	0.7	FALSE	0.089	0.342	0.574	0.604	2.413	False positive rate
0.5	A = 0.5	Delta_i = .25	14	0.7	TRUE	0.017	0.477	0.707	0.737	2.249	True positive rate
0.5	A = 0.5	Delta_i = .25	16	0.8	TRUE	0.010	0.502	0.754	0.770	2.250	True positive rate
0.5	A = 0.5	Delta_i = .25	16	0.8	FALSE	0.117	0.288	0.542	0.548	2.492	False positive rate
0.5	A = 0.5	Delta_i = .25	18	0.9	TRUE	0.005	0.500	0.772	0.775	2.268	True positive rate
0.5	A = 0.5	Delta_i = .25	18	0.9	FALSE	0.124	0.253	0.535	0.518	2.571	False positive rate
0.5	A = 0.5	Delta_i = .25	20	1.0	TRUE	0.004	0.504	0.809	0.789	2.298	True positive rate
0.5	A = 0.5	Delta_i = .50	0	0.0	FALSE	0.005	0.502	0.803	0.794	2.298	False positive rate
0.5	A = 0.5	Delta_i = .50	2	0.1	TRUE	0.469	0.079	0.230	0.196	2.793	True positive rate

slope_multiplier	slipier	cell	target_Delta_i	diff	diff	true_diff	lagged_mean	mean_p	adjusted	height	diagnostic
0.5	A =	Delta_i	2	0.1	FALSE	0.008	0.509	0.767	0.789	2.243	False positive rate
0.5	A =	Delta_i	4	0.2	FALSE	0.014	0.514	0.730	0.781	2.201	False positive rate
0.5	A =	Delta_i	4	0.2	TRUE	0.531	0.080	0.187	0.177	2.689	True positive rate
0.5	A =	Delta_i	6	0.3	TRUE	0.547	0.088	0.173	0.178	2.598	True positive rate
0.5	A =	Delta_i	6	0.3	FALSE	0.027	0.510	0.684	0.762	2.170	False positive rate
0.5	A =	Delta_i	8	0.4	TRUE	0.550	0.122	0.193	0.212	2.500	True positive rate
0.5	A =	Delta_i	8	0.4	FALSE	0.058	0.496	0.637	0.731	2.164	False positive rate
0.5	A =	Delta_i	10	0.5	FALSE	0.279	0.368	0.467	0.543	2.279	False positive rate
0.5	A =	Delta_i	10	0.5	TRUE	0.358	0.288	0.378	0.425	2.317	True positive rate
0.5	A =	Delta_i	12	0.6	TRUE	0.094	0.477	0.612	0.687	2.175	True positive rate
0.5	A =	Delta_i	12	0.6	FALSE	0.587	0.143	0.203	0.234	2.483	False positive rate
0.5	A =	Delta_i	14	0.7	TRUE	0.040	0.503	0.672	0.739	2.170	True positive rate
0.5	A =	Delta_i	14	0.7	FALSE	0.612	0.089	0.159	0.173	2.598	False positive rate

slope_multiplier	sloper_cell	target_Delta_i	diff	ip_diff	true_diff	lagged_mean	mean_p	adjusted	height	diagnostic
0.5	A = 0.5	Delta_i = .50	16	0.8	TRUE	0.018	0.516	0.724	0.766	2.194 True positive rate
0.5	A = 0.5	Delta_i = .50	16	0.8	FALSE	0.655	0.052	0.124	0.122	2.715 False positive rate
0.5	A = 0.5	Delta_i = .50	18	0.9	TRUE	0.012	0.507	0.756	0.772	2.238 True positive rate
0.5	A = 0.5	Delta_i = .50	18	0.9	FALSE	0.602	0.049	0.157	0.139	2.845 False positive rate
0.5	A = 0.5	Delta_i = .50	20	1.0	TRUE	0.006	0.503	0.809	0.785	2.298 True positive rate
1.0	A = 1	Delta_i = .25	0	0.0	FALSE	0.004	0.510	0.818	0.802	2.298 False positive rate
1.0	A = 1	Delta_i = .25	2	0.1	FALSE	0.006	0.502	0.776	0.788	2.267 False positive rate
1.0	A = 1	Delta_i = .25	2	0.1	TRUE	0.139	0.243	0.515	0.498	2.584 True positive rate
1.0	A = 1	Delta_i = .25	4	0.2	FALSE	0.010	0.495	0.738	0.775	2.246 False positive rate
1.0	A = 1	Delta_i = .25	4	0.2	TRUE	0.140	0.256	0.500	0.508	2.507 True positive rate
1.0	A = 1	Delta_i = .25	6	0.3	TRUE	0.131	0.295	0.508	0.540	2.437 True positive rate
1.0	A = 1	Delta_i = .25	6	0.3	FALSE	0.017	0.486	0.705	0.754	2.239 False positive rate
1.0	A = 1	Delta_i = .25	8	0.4	FALSE	0.034	0.462	0.671	0.726	2.258 False positive rate

slope_multiplier	slope	cell	target_Delta_i	diff	true_diff	lagged_mean	mean_p	adjusted	height	diagnostic	
1.0	A =	Delta_i	8	0.4	TRUE	0.098	0.351	0.558	0.601	2.358	True positive rate
	1	= .25									
1.0	A =	Delta_i	10	0.5	TRUE	0.068	0.400	0.596	0.652	2.299	True positive rate
	1	= .25									
1.0	A =	Delta_i	10	0.5	FALSE	0.069	0.404	0.601	0.660	2.298	False positive rate
	1	= .25									
1.0	A =	Delta_i	12	0.6	FALSE	0.106	0.344	0.543	0.594	2.367	False positive rate
	1	= .25									
1.0	A =	Delta_i	12	0.6	TRUE	0.039	0.452	0.655	0.713	2.253	True positive rate
	1	= .25									
1.0	A =	Delta_i	14	0.7	TRUE	0.016	0.487	0.709	0.757	2.239	True positive rate
	1	= .25									
1.0	A =	Delta_i	14	0.7	FALSE	0.102	0.303	0.530	0.561	2.437	False positive rate
	1	= .25									
1.0	A =	Delta_i	16	0.8	TRUE	0.015	0.495	0.740	0.773	2.248	True positive rate
	1	= .25									
1.0	A =	Delta_i	16	0.8	FALSE	0.116	0.275	0.527	0.534	2.499	False positive rate
	1	= .25									
1.0	A =	Delta_i	18	0.9	FALSE	0.121	0.233	0.520	0.496	2.604	False positive rate
	1	= .25									
1.0	A =	Delta_i	18	0.9	TRUE	0.005	0.510	0.787	0.796	2.264	True positive rate
	1	= .25									
1.0	A =	Delta_i	20	1.0	TRUE	0.005	0.511	0.821	0.804	2.298	True positive rate
	1	= .25									
1.0	A =	Delta_i	0	0.0	FALSE	0.004	0.502	0.806	0.795	2.298	False positive rate
	1	= .50									

slope_multiplier	slope	cell	target_Delta	diff	true_diff	flagged	mean	mean_p	adjusted	height	diagnostic
1.0	A =	Delta_i	2	0.1	FALSE	0.008	0.510	0.761	0.785	2.237	False positive rate
	1	= .50									
1.0	A =	Delta_i	2	0.1	TRUE	0.620	0.041	0.142	0.122	2.846	True positive rate
	1	= .50									
1.0	A =	Delta_i	4	0.2	FALSE	0.016	0.515	0.724	0.778	2.193	False positive rate
	1	= .50									
1.0	A =	Delta_i	4	0.2	TRUE	0.688	0.047	0.114	0.110	2.721	True positive rate
	1	= .50									
1.0	A =	Delta_i	6	0.3	FALSE	0.033	0.507	0.672	0.754	2.160	False positive rate
	1	= .50									
1.0	A =	Delta_i	6	0.3	TRUE	0.701	0.057	0.109	0.118	2.620	True positive rate
	1	= .50									
1.0	A =	Delta_i	8	0.4	TRUE	0.665	0.098	0.146	0.168	2.514	True positive rate
	1	= .50									
1.0	A =	Delta_i	8	0.4	FALSE	0.078	0.485	0.615	0.715	2.154	False positive rate
	1	= .50									
1.0	A =	Delta_i	10	0.5	FALSE	0.350	0.321	0.404	0.474	2.297	False positive rate
	1	= .50									
1.0	A =	Delta_i	10	0.5	TRUE	0.387	0.303	0.383	0.445	2.299	True positive rate
	1	= .50									
1.0	A =	Delta_i	12	0.6	TRUE	0.087	0.481	0.613	0.708	2.159	True positive rate
	1	= .50									
1.0	A =	Delta_i	12	0.6	FALSE	0.639	0.114	0.165	0.190	2.508	False positive rate
	1	= .50									
1.0	A =	Delta_i	14	0.7	TRUE	0.023	0.515	0.686	0.767	2.158	True positive rate
	1	= .50									

slope_multiplier	slope	cell	target_Delta_i	dip	diff	true_diff	lagged_mean	mean_p	adjusted	height	diagnostic
1.0	A =	Delta_i	14	0.7	FALSE	0.698	0.053	0.106	0.112	2.625	False positive rate
1.0	1	= .50									
1.0	A =	Delta_i	16	0.8	TRUE	0.019	0.510	0.717	0.772	2.193	True positive rate
1.0	1	= .50									
1.0	A =	Delta_i	16	0.8	FALSE	0.671	0.047	0.119	0.117	2.720	False positive rate
1.0	1	= .50									
1.0	A =	Delta_i	18	0.9	TRUE	0.008	0.508	0.761	0.785	2.237	True positive rate
1.0	1	= .50									
1.0	A =	Delta_i	18	0.9	FALSE	0.651	0.041	0.139	0.125	2.851	False positive rate
1.0	1	= .50									
1.0	A =	Delta_i	20	1.0	TRUE	0.004	0.508	0.818	0.800	2.298	True positive rate
1.0	1	= .50									
1.5	A =	Delta_i	0	0.0	FALSE	0.004	0.507	0.813	0.801	2.298	False positive rate
1.5	1.5	= .25									
1.5	A =	Delta_i	2	0.1	FALSE	0.010	0.500	0.762	0.781	2.260	False positive rate
1.5	1.5	= .25									
1.5	A =	Delta_i	2	0.1	TRUE	0.210	0.172	0.419	0.421	2.647	True positive rate
1.5	1.5	= .25									
1.5	A =	Delta_i	4	0.2	FALSE	0.013	0.498	0.733	0.773	2.240	False positive rate
1.5	1.5	= .25									
1.5	A =	Delta_i	4	0.2	TRUE	0.188	0.233	0.454	0.484	2.533	True positive rate
1.5	1.5	= .25									
1.5	A =	Delta_i	6	0.3	FALSE	0.023	0.473	0.682	0.739	2.233	False positive rate
1.5	1.5	= .25									
1.5	A =	Delta_i	6	0.3	TRUE	0.171	0.275	0.470	0.524	2.451	True positive rate
1.5	1.5	= .25									

slope_multiplier	slope	cell	target_Delta_i	diff	diff	true_diff	lagged_mean	mean_p	adjusted	height	diagnostic
1.5	A =	Delta_i	8	0.4	TRUE	0.125	0.345	0.529	0.602	2.356	True positive rate
1.5	1.5	= .25									
1.5	A =	Delta_i	8	0.4	FALSE	0.051	0.432	0.624	0.686	2.260	False positive rate
1.5	1.5	= .25									
1.5	A =	Delta_i	10	0.5	TRUE	0.078	0.414	0.600	0.681	2.284	True positive rate
1.5	1.5	= .25									
1.5	A =	Delta_i	10	0.5	FALSE	0.083	0.371	0.556	0.616	2.313	False positive rate
1.5	1.5	= .25									
1.5	A =	Delta_i	12	0.6	FALSE	0.123	0.297	0.489	0.533	2.392	False positive rate
1.5	1.5	= .25									
1.5	A =	Delta_i	12	0.6	TRUE	0.033	0.473	0.668	0.752	2.236	True positive rate
1.5	1.5	= .25									
1.5	A =	Delta_i	14	0.7	TRUE	0.016	0.490	0.705	0.773	2.228	True positive rate
1.5	1.5	= .25									
1.5	A =	Delta_i	14	0.7	FALSE	0.127	0.267	0.483	0.515	2.463	False positive rate
1.5	1.5	= .25									
1.5	A =	Delta_i	16	0.8	TRUE	0.006	0.500	0.743	0.791	2.239	True positive rate
1.5	1.5	= .25									
1.5	A =	Delta_i	16	0.8	FALSE	0.120	0.244	0.491	0.498	2.536	False positive rate
1.5	1.5	= .25									
1.5	A =	Delta_i	18	0.9	TRUE	0.006	0.501	0.774	0.801	2.262	True positive rate
1.5	1.5	= .25									
1.5	A =	Delta_i	18	0.9	FALSE	0.108	0.220	0.506	0.487	2.624	False positive rate
1.5	1.5	= .25									
1.5	A =	Delta_i	20	1.0	TRUE	0.003	0.508	0.813	0.812	2.298	True positive rate
1.5	1.5	= .25									

slope_multiplier	slipier	cell	target_Delta_i	diff	truediff	flagged	mean	mean_p	adjusted	height	diagnostic
1.5	A =	Delta_i	0	0.0	FALSE	0.004	0.500	0.804	0.795	2.298	False positive rate
1.5	A =	Delta_i	2	0.1	FALSE	0.009	0.506	0.756	0.782	2.234	False positive rate
1.5	A =	Delta_i	2	0.1	TRUE	0.758	0.021	0.080	0.071	2.877	True positive rate
1.5	A =	Delta_i	4	0.2	FALSE	0.017	0.503	0.701	0.765	2.187	False positive rate
1.5	A =	Delta_i	4	0.2	TRUE	0.835	0.022	0.054	0.055	2.744	True positive rate
1.5	A =	Delta_i	6	0.3	FALSE	0.026	0.516	0.679	0.764	2.149	False positive rate
1.5	A =	Delta_i	6	0.3	TRUE	0.835	0.023	0.048	0.052	2.646	True positive rate
1.5	A =	Delta_i	8	0.4	TRUE	0.753	0.086	0.118	0.139	2.523	True positive rate
1.5	A =	Delta_i	8	0.4	FALSE	0.089	0.492	0.622	0.716	2.149	False positive rate
1.5	A =	Delta_i	10	0.5	FALSE	0.442	0.261	0.331	0.386	2.331	False positive rate
1.5	A =	Delta_i	10	0.5	TRUE	0.341	0.349	0.431	0.516	2.266	True positive rate
1.5	A =	Delta_i	12	0.6	FALSE	0.706	0.069	0.111	0.126	2.542	False positive rate
1.5	A =	Delta_i	12	0.6	TRUE	0.048	0.502	0.637	0.755	2.136	True positive rate

slope_multiplier	slope	cell	target_Delta_i	n_dif	p_dif	true_dif	flagged_rate	mean	mean_p	adjusted_rate	height	diagnostic
1.5	A =	Delta_i	14	0.7	TRUE	0.026	0.507	0.671	0.770	2.153	True positive rate	
1.5	1.5	= .50										
1.5	A =	Delta_i	14	0.7	FALSE	0.712	0.044	0.092	0.097	2.637	False positive rate	
1.5	1.5	= .50										
1.5	A =	Delta_i	16	0.8	TRUE	0.015	0.510	0.715	0.785	2.189	True positive rate	
1.5	1.5	= .50										
1.5	A =	Delta_i	16	0.8	FALSE	0.702	0.039	0.098	0.097	2.735	False positive rate	
1.5	1.5	= .50										
1.5	A =	Delta_i	18	0.9	TRUE	0.009	0.507	0.759	0.795	2.236	True positive rate	
1.5	1.5	= .50										
1.5	A =	Delta_i	18	0.9	FALSE	0.624	0.043	0.135	0.122	2.855	False positive rate	
1.5	1.5	= .50										
1.5	A =	Delta_i	20	1.0	TRUE	0.004	0.505	0.810	0.809	2.298	True positive rate	
1.5	1.5	= .50										

```

item_diag_plots <- plot_by_effect(item_diag, function(plot_data) {
  ggplot(
    plot_data,
    aes(x = n_dif, y = flagged_rate, color = diagnostic, group = diagnostic,
        shape = p_dif >= 0.5)) +
    geom_rect(
      data = breakdown_band(plot_data),
      aes(xmin = xmin, xmax = xmax, ymin = ymin, ymax = ymax),
      inherit.aes = FALSE,
      fill = "#B85C38",
      alpha = 0.08) +
    geom_hline(yintercept = alpha, linetype = 3, color = "grey50") +
    geom_point(size = 2.1, na.rm = TRUE) +
    geom_line(linewidth = 0.75, na.rm = TRUE) +
    scale_y_continuous(limits = c(0, 1), breaks = seq(0, 1, 0.25)) +
    scale_color_manual(
      values = c("False positive rate" = "grey50",
        "True positive rate" = "#7BAFD4")) +

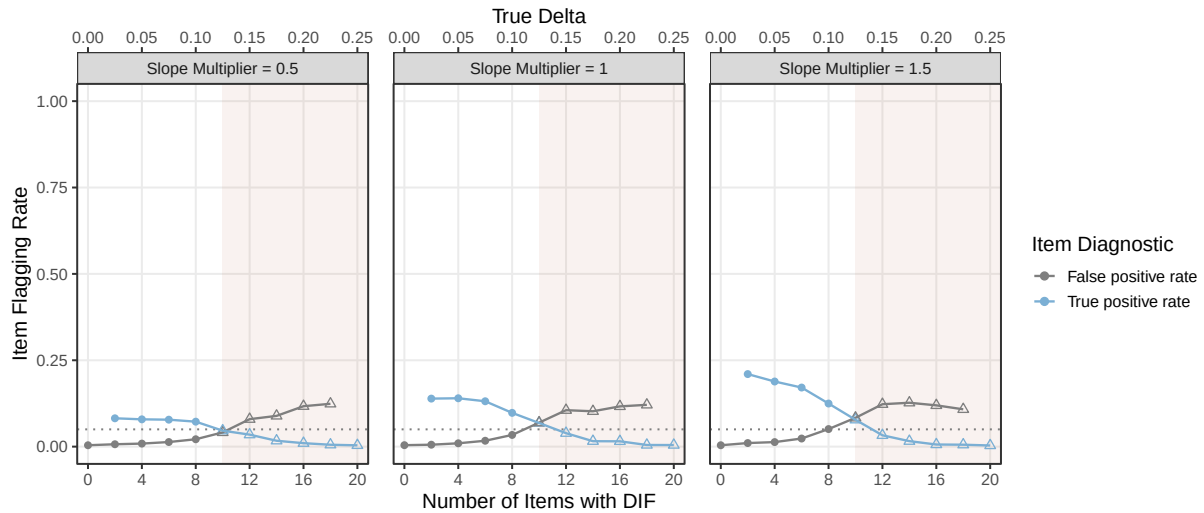
```

```

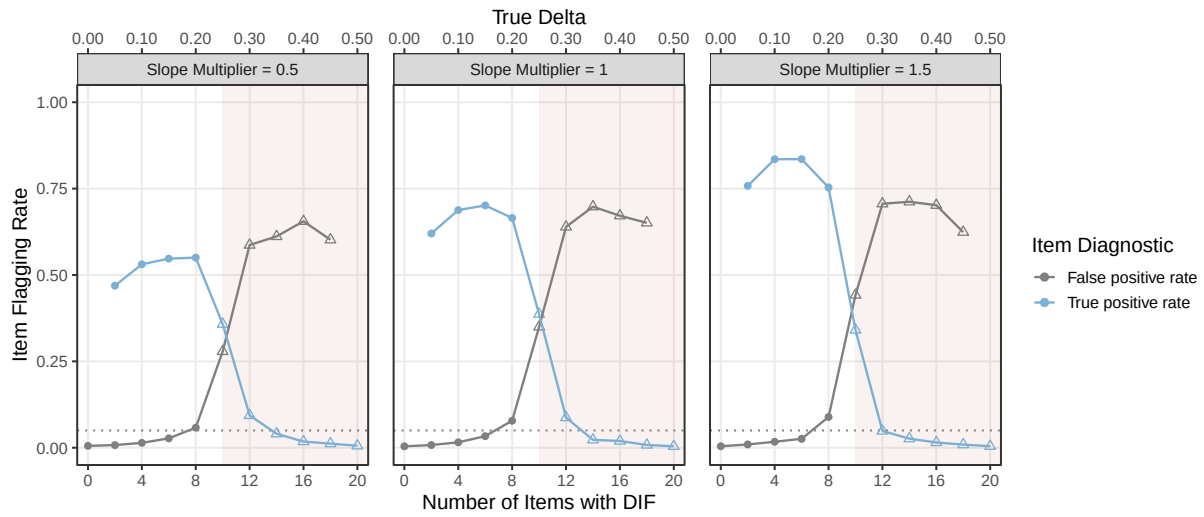
scale_shape_manual(
  values = c("FALSE" = 16, "TRUE" = 2),
  guide = "none") +
ndif_x_scale(plot_data) +
coord_cartesian(clip = "off") +
sim2_public_facets() +
labs(
  x = "Number of Items with DIF",
  y = "Item Flagging Rate",
  color = "Item Diagnostic") +
public_plot_theme
})

```

Item-Level Effect = 0.25



Item-Level Effect = 0.50



```
item_weight_summary <- item_diag |>
  select(
    slope_multiplier, target_Delta_i, n_dif, true_dif,
    mean_weight, mean_k) |>
  mutate(item_type = ifelse(true_dif, "True DIF", "True Anchor"))

knitr::kable(item_weight_summary, digits = 3)
```

slope_multiplier	target_Delta_i	n_dif	true_dif	mean_weight	mean_k	item_type
0.5	Delta_i = .25	0	FALSE	0.797	2.298	True Anchor
0.5	Delta_i = .25	2	FALSE	0.789	2.273	True Anchor
0.5	Delta_i = .25	2	TRUE	0.548	2.530	True DIF
0.5	Delta_i = .25	4	FALSE	0.782	2.255	True Anchor
0.5	Delta_i = .25	4	TRUE	0.566	2.472	True DIF
0.5	Delta_i = .25	6	TRUE	0.581	2.423	True DIF
0.5	Delta_i = .25	6	FALSE	0.773	2.245	True Anchor
0.5	Delta_i = .25	8	TRUE	0.603	2.374	True DIF
0.5	Delta_i = .25	8	FALSE	0.756	2.248	True Anchor
0.5	Delta_i = .25	10	FALSE	0.712	2.285	True Anchor

slope_multiplier	target_Delta_i	n_dif	true_dif	mean_weight	mean_k	item_type
0.5	Delta_i = .25	10	TRUE	0.652	2.311	True DIF
0.5	Delta_i = .25	12	TRUE	0.697	2.265	True DIF
0.5	Delta_i = .25	12	FALSE	0.645	2.348	True Anchor
0.5	Delta_i = .25	14	FALSE	0.604	2.413	True Anchor
0.5	Delta_i = .25	14	TRUE	0.737	2.249	True DIF
0.5	Delta_i = .25	16	TRUE	0.770	2.250	True DIF
0.5	Delta_i = .25	16	FALSE	0.548	2.492	True Anchor
0.5	Delta_i = .25	18	TRUE	0.775	2.268	True DIF
0.5	Delta_i = .25	18	FALSE	0.518	2.571	True Anchor
0.5	Delta_i = .25	20	TRUE	0.789	2.298	True DIF
0.5	Delta_i = .50	0	FALSE	0.794	2.298	True Anchor
0.5	Delta_i = .50	2	TRUE	0.196	2.793	True DIF
0.5	Delta_i = .50	2	FALSE	0.789	2.243	True Anchor
0.5	Delta_i = .50	4	FALSE	0.781	2.201	True Anchor
0.5	Delta_i = .50	4	TRUE	0.177	2.689	True DIF
0.5	Delta_i = .50	6	TRUE	0.178	2.598	True DIF
0.5	Delta_i = .50	6	FALSE	0.762	2.170	True Anchor
0.5	Delta_i = .50	8	TRUE	0.212	2.500	True DIF
0.5	Delta_i = .50	8	FALSE	0.731	2.164	True Anchor
0.5	Delta_i = .50	10	FALSE	0.543	2.279	True Anchor
0.5	Delta_i = .50	10	TRUE	0.425	2.317	True DIF
0.5	Delta_i = .50	12	TRUE	0.687	2.175	True DIF
0.5	Delta_i = .50	12	FALSE	0.234	2.483	True Anchor
0.5	Delta_i = .50	14	TRUE	0.739	2.170	True DIF
0.5	Delta_i = .50	14	FALSE	0.173	2.598	True Anchor
0.5	Delta_i = .50	16	TRUE	0.766	2.194	True DIF
0.5	Delta_i = .50	16	FALSE	0.122	2.715	True Anchor
0.5	Delta_i = .50	18	TRUE	0.772	2.238	True DIF

slope_multiplier	target_Delta_i	n_dif	true_dif	mean_weight	mean_k	item_type
0.5	Delta_i = .50	18	FALSE	0.139	2.845	True Anchor
0.5	Delta_i = .50	20	TRUE	0.785	2.298	True DIF
1.0	Delta_i = .25	0	FALSE	0.802	2.298	True Anchor
1.0	Delta_i = .25	2	FALSE	0.788	2.267	True Anchor
1.0	Delta_i = .25	2	TRUE	0.498	2.584	True DIF
1.0	Delta_i = .25	4	FALSE	0.775	2.246	True Anchor
1.0	Delta_i = .25	4	TRUE	0.508	2.507	True DIF
1.0	Delta_i = .25	6	TRUE	0.540	2.437	True DIF
1.0	Delta_i = .25	6	FALSE	0.754	2.239	True Anchor
1.0	Delta_i = .25	8	FALSE	0.726	2.258	True Anchor
1.0	Delta_i = .25	8	TRUE	0.601	2.358	True DIF
1.0	Delta_i = .25	10	TRUE	0.652	2.299	True DIF
1.0	Delta_i = .25	10	FALSE	0.660	2.298	True Anchor
1.0	Delta_i = .25	12	FALSE	0.594	2.367	True Anchor
1.0	Delta_i = .25	12	TRUE	0.713	2.253	True DIF
1.0	Delta_i = .25	14	TRUE	0.757	2.239	True DIF
1.0	Delta_i = .25	14	FALSE	0.561	2.437	True Anchor
1.0	Delta_i = .25	16	TRUE	0.773	2.248	True DIF
1.0	Delta_i = .25	16	FALSE	0.534	2.499	True Anchor
1.0	Delta_i = .25	18	FALSE	0.496	2.604	True Anchor
1.0	Delta_i = .25	18	TRUE	0.796	2.264	True DIF
1.0	Delta_i = .25	20	TRUE	0.804	2.298	True DIF
1.0	Delta_i = .50	0	FALSE	0.795	2.298	True Anchor
1.0	Delta_i = .50	2	FALSE	0.785	2.237	True Anchor
1.0	Delta_i = .50	2	TRUE	0.122	2.846	True DIF
1.0	Delta_i = .50	4	FALSE	0.778	2.193	True Anchor
1.0	Delta_i = .50	4	TRUE	0.110	2.721	True DIF

slope_multiplier	target_Delta_i	n_dif	true_dif	mean_weight	mean_k	item_type
1.0	Delta_i = .50	6	FALSE	0.754	2.160	True Anchor
1.0	Delta_i = .50	6	TRUE	0.118	2.620	True DIF
1.0	Delta_i = .50	8	TRUE	0.168	2.514	True DIF
1.0	Delta_i = .50	8	FALSE	0.715	2.154	True Anchor
1.0	Delta_i = .50	10	FALSE	0.474	2.297	True Anchor
1.0	Delta_i = .50	10	TRUE	0.445	2.299	True DIF
1.0	Delta_i = .50	12	TRUE	0.708	2.159	True DIF
1.0	Delta_i = .50	12	FALSE	0.190	2.508	True Anchor
1.0	Delta_i = .50	14	TRUE	0.767	2.158	True DIF
1.0	Delta_i = .50	14	FALSE	0.112	2.625	True Anchor
1.0	Delta_i = .50	16	TRUE	0.772	2.193	True DIF
1.0	Delta_i = .50	16	FALSE	0.117	2.720	True Anchor
1.0	Delta_i = .50	18	TRUE	0.785	2.237	True DIF
1.0	Delta_i = .50	18	FALSE	0.125	2.851	True Anchor
1.0	Delta_i = .50	20	TRUE	0.800	2.298	True DIF
1.5	Delta_i = .25	0	FALSE	0.801	2.298	True Anchor
1.5	Delta_i = .25	2	FALSE	0.781	2.260	True Anchor
1.5	Delta_i = .25	2	TRUE	0.421	2.647	True DIF
1.5	Delta_i = .25	4	FALSE	0.773	2.240	True Anchor
1.5	Delta_i = .25	4	TRUE	0.484	2.533	True DIF
1.5	Delta_i = .25	6	FALSE	0.739	2.233	True Anchor
1.5	Delta_i = .25	6	TRUE	0.524	2.451	True DIF
1.5	Delta_i = .25	8	TRUE	0.602	2.356	True DIF
1.5	Delta_i = .25	8	FALSE	0.686	2.260	True Anchor
1.5	Delta_i = .25	10	TRUE	0.681	2.284	True DIF
1.5	Delta_i = .25	10	FALSE	0.616	2.313	True Anchor
1.5	Delta_i = .25	12	FALSE	0.533	2.392	True Anchor

slope_multiplier	target_Delta_i	n_dif	true_dif	mean_weight	mean_k	item_type
1.5	Delta_i = .25	12	TRUE	0.752	2.236	True DIF
1.5	Delta_i = .25	14	TRUE	0.773	2.228	True DIF
1.5	Delta_i = .25	14	FALSE	0.515	2.463	True Anchor
1.5	Delta_i = .25	16	TRUE	0.791	2.239	True DIF
1.5	Delta_i = .25	16	FALSE	0.498	2.536	True Anchor
1.5	Delta_i = .25	18	TRUE	0.801	2.262	True DIF
1.5	Delta_i = .25	18	FALSE	0.487	2.624	True Anchor
1.5	Delta_i = .25	20	TRUE	0.812	2.298	True DIF
1.5	Delta_i = .50	0	FALSE	0.795	2.298	True Anchor
1.5	Delta_i = .50	2	FALSE	0.782	2.234	True Anchor
1.5	Delta_i = .50	2	TRUE	0.071	2.877	True DIF
1.5	Delta_i = .50	4	FALSE	0.765	2.187	True Anchor
1.5	Delta_i = .50	4	TRUE	0.055	2.744	True DIF
1.5	Delta_i = .50	6	FALSE	0.764	2.149	True Anchor
1.5	Delta_i = .50	6	TRUE	0.052	2.646	True DIF
1.5	Delta_i = .50	8	TRUE	0.139	2.523	True DIF
1.5	Delta_i = .50	8	FALSE	0.716	2.149	True Anchor
1.5	Delta_i = .50	10	FALSE	0.386	2.331	True Anchor
1.5	Delta_i = .50	10	TRUE	0.516	2.266	True DIF
1.5	Delta_i = .50	12	FALSE	0.126	2.542	True Anchor
1.5	Delta_i = .50	12	TRUE	0.755	2.136	True DIF
1.5	Delta_i = .50	14	TRUE	0.770	2.153	True DIF
1.5	Delta_i = .50	14	FALSE	0.097	2.637	True Anchor
1.5	Delta_i = .50	16	TRUE	0.785	2.189	True DIF
1.5	Delta_i = .50	16	FALSE	0.097	2.735	True Anchor
1.5	Delta_i = .50	18	TRUE	0.795	2.236	True DIF
1.5	Delta_i = .50	18	FALSE	0.122	2.855	True Anchor

Supplementary Analysis of Item-Level Variances

This diagnostic isolates the relationship between item discrimination and the standard error of the item-level scaling function. There is no DIF in this supplementary setup: the target item has the same discrimination and intercept in both groups, and the latent distributions are identical. The target item's discrimination is varied from $a = 0.5$ to $a = 2.0$ in both groups.

```
item_variance_diag_file <- file.path(
  results_dir,
  "sim2_item_level_variance_diagnostic.rds")

run_item_level_variance_diagnostic <- function(
  a.grid = seq(0.5, 2, by = 0.1),
  n.reps = 100,
  n.persons = 500,
  n.items = 20,
  focal.item = 1,
  fun = "d_fun3",
  seed = 20260728,
  n.cores = NULL) {

  stopifnot(focal.item >= 1, focal.item <= n.items)
  check_archive_robustDIF()
  if (is.null(n.cores) || is.na(n.cores)) {
    detected.cores <- parallel::detectCores()
    n.cores <- if (is.na(detected.cores)) 1 else max(1, min(10, detected.cores - 2))
  }

  tasks <- expand.grid(
    focal_a = a.grid,
    rep = seq_len(n.reps),
    KEEP.OUT.ATTRS = FALSE)
  tasks$row_id <- seq_len(nrow(tasks))

  base.a <- seq(0.75, 1.75, length.out = n.items)
  base.b <- seq(-1.8, 1.8, length.out = n.items)
  base.b[focal.item] <- 0
```

```

run_task <- function(i) {
  set.seed(seed + tasks$row_id[i])

  a <- base.a
  a[focal.item] <- tasks$focal_a[i]
  d <- -base.b * a

  theta0 <- rnorm(n.persons)
  theta1 <- rnorm(n.persons)
  dat0 <- simdata(a, d, n.persons, "2PL", Theta = matrix(theta0))
  dat1 <- simdata(a, d, n.persons, "2PL", Theta = matrix(theta1))

  out <- tryCatch({
    fit0 <- mirt(dat0, 1, SE = TRUE, verbose = FALSE)
    fit1 <- mirt(dat1, 1, SE = TRUE, verbose = FALSE)
    mle <- robustDIF::get_model_parms(list(fit0, fit1))
    y <- robustDIF::y_fun(mle, fun)
    y.se <- sqrt(Matrix::diag(robustDIF::vcov_y(mle, fun = fun)))

    data.frame(
      focal_a = tasks$focal_a[i],
      rep = tasks$rep[i],
      item = focal.item,
      y_hat = y[focal.item],
      y_se = y.se[focal.item],
      converged_0 = isTRUE(fit0@OptimInfo$converged),
      converged_1 = isTRUE(fit1@OptimInfo$converged),
      error = NA_character_)
  }, error = function(e) {
    data.frame(
      focal_a = tasks$focal_a[i],
      rep = tasks$rep[i],
      item = focal.item,
      y_hat = NA_real_,
      y_se = NA_real_,
      converged_0 = FALSE,
      converged_1 = FALSE,
      error = conditionMessage(e))
  })

  out
}

```

```

diag_rows <- parallel::mclapply(
  seq_len(nrow(tasks)),
  run_task,
  mc.cores = n.cores,
  mc.set.seed = FALSE)

bind_rows(diag_rows)
}

if (file.exists(item_variance_diag_file)) {
  item_variance_diag <- readRDS(item_variance_diag_file)
} else {
  item_variance_diag <- run_item_level_variance_diagnostic()
  saveRDS(item_variance_diag, item_variance_diag_file)
}

item_variance_diag_summary <- item_variance_diag |>
  summarize(
    mean_y = mean_or_na(y_hat),
    empirical_sd_y = sd_or_na(y_hat),
    mean_se_y = mean_or_na(y_se),
    q10_se_y = quantile(y_se, 0.10, na.rm = TRUE),
    q90_se_y = quantile(y_se, 0.90, na.rm = TRUE),
    failure_rate = mean(is.na(y_se) | !converged_0 | !converged_1),
    .by = focal_a)

knitr::kable(item_variance_diag_summary, digits = 3)

```

focal_a	mean_y	empirical_sd_y	mean_se_y	q10_se_y	q90_se_y	failure_rate
0.5	-0.016	0.288	0.266	0.219	0.325	0
0.6	-0.001	0.223	0.230	0.194	0.263	0
0.7	0.000	0.186	0.200	0.182	0.218	0
0.8	-0.005	0.197	0.182	0.162	0.206	0
0.9	-0.012	0.172	0.166	0.151	0.182	0
1.0	0.000	0.143	0.154	0.142	0.165	0
1.1	-0.008	0.151	0.143	0.131	0.155	0
1.2	-0.005	0.138	0.135	0.125	0.145	0
1.3	0.014	0.141	0.129	0.120	0.139	0
1.4	-0.001	0.126	0.123	0.115	0.130	0
1.5	-0.021	0.118	0.119	0.111	0.126	0

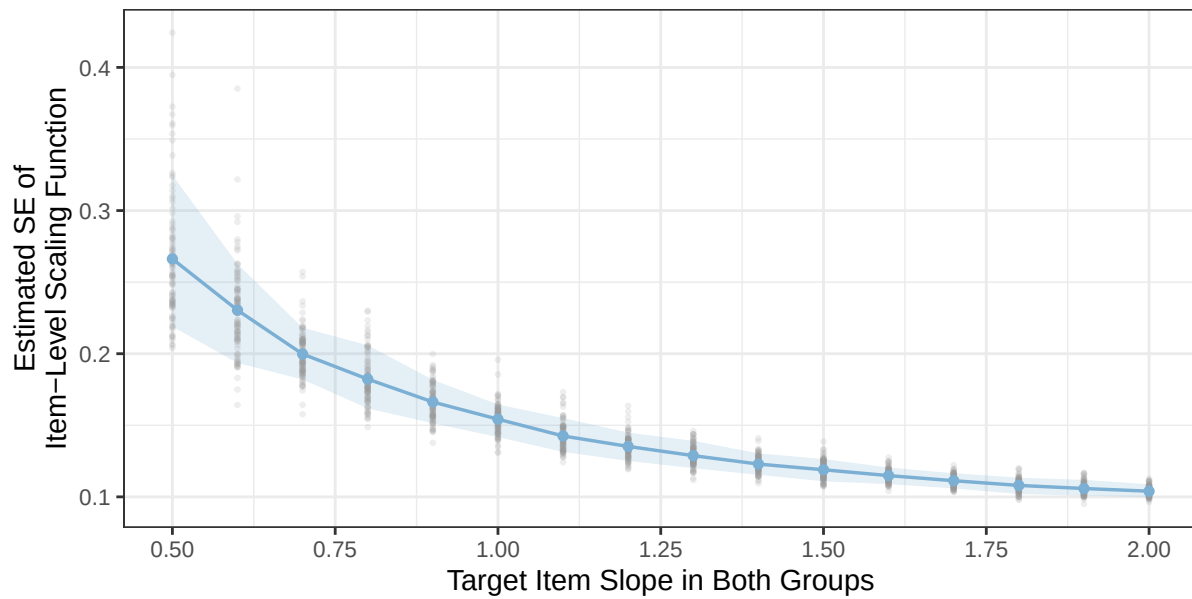
focal_a	mean_y	empirical_sd_y	mean_se_y	q10_se_y	q90_se_y	failure_rate
1.6	0.007	0.115	0.115	0.109	0.120	0
1.7	0.011	0.119	0.111	0.106	0.117	0
1.8	0.009	0.105	0.108	0.102	0.113	0
1.9	0.005	0.119	0.106	0.101	0.112	0
2.0	-0.005	0.112	0.104	0.100	0.109	0

```

item_variance_plot <- ggplot(
  item_variance_diag_summary,
  aes(x = focal_a, y = mean_se_y)) +
  geom_ribbon(
    aes(ymin = q10_se_y, ymax = q90_se_y),
    fill = "#7BAFD4",
    alpha = 0.20) +
  geom_point(
    data = item_variance_diag,
    aes(x = focal_a, y = y_se),
    inherit.aes = FALSE,
    color = "grey60",
    alpha = 0.16,
    size = 0.8,
    na.rm = TRUE) +
  geom_line(color = "#7BAFD4", linewidth = 1) +
  geom_point(color = "#7BAFD4", size = 2.2) +
  scale_x_continuous(breaks = seq(0.5, 2, by = 0.25)) +
  labs(
    x = "Target Item Slope in Both Groups",
    y = "Estimated SE of\nItem-Level Scaling Function") +
  theme_bw(base_size = 18) +
  theme(plot.margin = margin(t = 12, r = 12, b = 12, l = 80))

item_variance_plot

```



```
ggsave(
  file.path(figure_dir, "item_level_variance_by_discrimination.png"),
  item_variance_plot,
  width = 8.5,
  height = 5.0,
  dpi = 200)
```

Bias and Precision

Scaling Function Bias

This figure parallels the scaling-function bias plot in Simulation 1. **Naive** is the mean of all item-level scaling functions. **RDIF** is the robust estimate of the common-item scaling function, and **True** is the oracle unit-weighted estimate using the known anchor set.

```
scale_bias <- bind_rows(
  sim_delta |>
  filter(method == "RDIF") |>
  transmute(
    Method = "Naive",
    slope_multiplier,
    slope_cell,
    slope_cell_public,
```

```

    target_Delta_i,
    target_Delta_i_num,
    target_Delta_i_public,
    n_dif,
    p_dif,
    true_Delta,
    Bias = naive_est - delta_0),
sim_delta |>
  transmute(
    Method = as.character(method),
    slope_multiplier,
    slope_cell,
    slope_cell_public,
    target_Delta_i,
    target_Delta_i_num,
    target_Delta_i_public,
    n_dif,
    p_dif,
    true_Delta,
    Bias = scale_est - delta_0)) |>
mutate(
  Method = factor(
    Method,
    levels = c("Naive", method_levels),
    ordered = TRUE),
  breakdown = p_dif >= 0.5)

scale_cols <- c("Naive" = "grey80", method_cols)

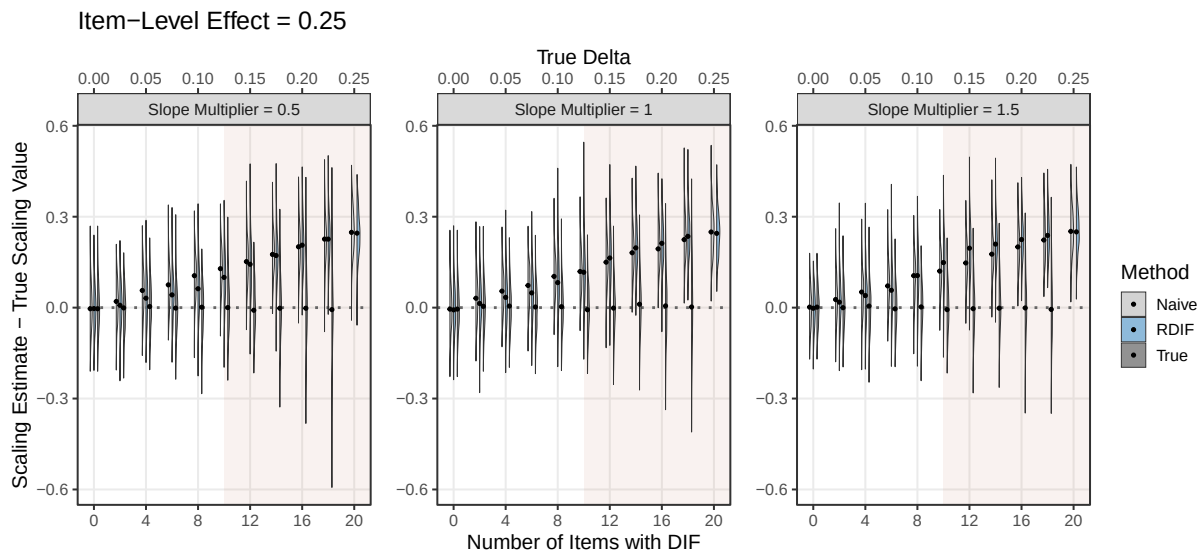
scale_bias_plots <- plot_by_effect(scale_bias, function(plot_data) {
  ggplot(plot_data, aes(x = n_dif, y = Bias, fill = Method)) +
    geom_rect(
      data = breakdown_band(plot_data),
      aes(xmin = xmin, xmax = xmax, ymin = ymin, ymax = ymax),
      inherit.aes = FALSE,
      fill = "#B85C38",
      alpha = 0.08) +
    geom_hline(yintercept = 0, linetype = 3, linewidth = 0.9, color = "grey40") +
    see::geom_violinhalf(
      aes(group = interaction(n_dif, Method)),
      position = position_dodge(width = 0.85),
      trim = TRUE,

```

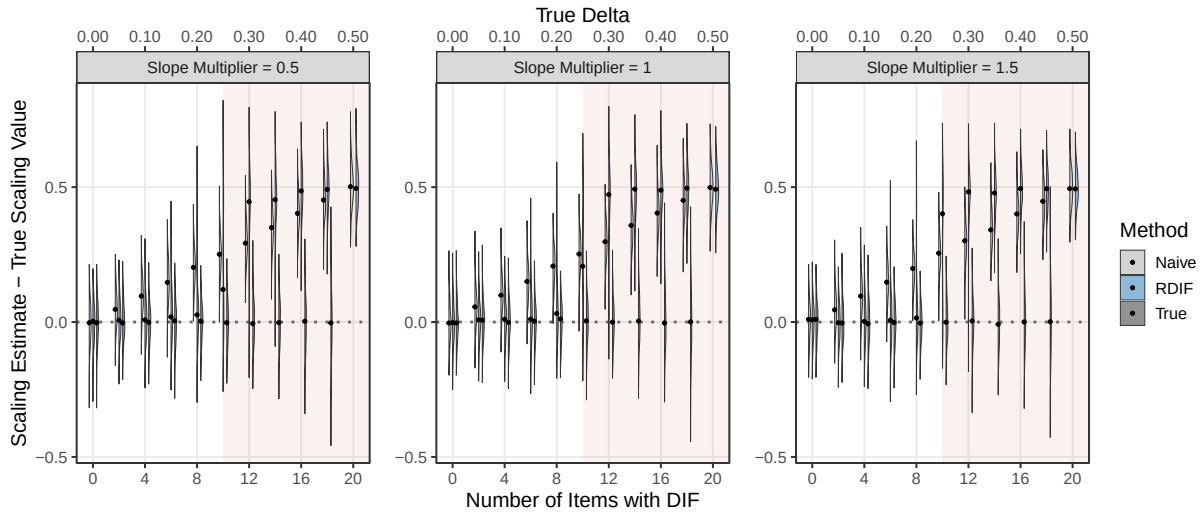
```

width = 2,
linewidth = 0.25,
alpha = 0.85) +
stat_summary(
  aes(group = interaction(n_dif, Method)),
  fun = median,
  geom = "point",
  position = position_dodge(width = 0.85),
  size = 0.8,
  color = "black") +
scale_fill_manual(values = scale_cols) +
ndif_x_scale(plot_data) +
coord_cartesian(clip = "off") +
sim2_public_facets(free_y = TRUE) +
labs(
  x = "Number of Items with DIF",
  y = "Scaling Estimate - True Scaling Value",
  fill = "Method") +
public_plot_theme
})

```



Item-Level Effect = 0.50



```
scale_bias_table <- scale_bias |>
  summarize(
    bias = mean_or_na(Bias),
    .by = c(Method, slope_multiplier, target_Delta_i, n_dif)) |>
  mutate(Method = as.character(Method)) |>
  arrange(slope_multiplier, target_Delta_i, n_dif, Method) |>
  pivot_wider(names_from = Method, values_from = bias)

knitr::kable(scale_bias_table, digits = 3)
```

slope_multiplier	target_Delta_i	n_dif	Naive	RDIF	True
0.5	Delta_i = .25	0	-0.002	-0.003	-0.002
0.5	Delta_i = .25	2	0.020	0.006	-0.004
0.5	Delta_i = .25	4	0.056	0.033	0.007
0.5	Delta_i = .25	6	0.073	0.042	-0.002
0.5	Delta_i = .25	8	0.101	0.065	-0.001
0.5	Delta_i = .25	10	0.129	0.102	0.005
0.5	Delta_i = .25	12	0.151	0.141	-0.002
0.5	Delta_i = .25	14	0.177	0.172	-0.003
0.5	Delta_i = .25	16	0.204	0.209	-0.003
0.5	Delta_i = .25	18	0.224	0.225	-0.004
0.5	Delta_i = .25	20	0.252	0.246	NA
0.5	Delta_i = .50	0	0.002	0.002	0.002
0.5	Delta_i = .50	2	0.053	0.007	0.003
0.5	Delta_i = .50	4	0.099	0.010	-0.003

slope_multiplier	target_Delta_i	n_dif	Naive	RDIF	True
0.5	Delta_i = .50	6	0.154	0.024	0.004
0.5	Delta_i = .50	8	0.200	0.046	-0.003
0.5	Delta_i = .50	10	0.249	0.195	-0.003
0.5	Delta_i = .50	12	0.297	0.403	-0.001
0.5	Delta_i = .50	14	0.347	0.446	-0.004
0.5	Delta_i = .50	16	0.407	0.489	0.000
0.5	Delta_i = .50	18	0.457	0.490	-0.007
0.5	Delta_i = .50	20	0.504	0.497	NA
1.0	Delta_i = .25	0	-0.003	-0.004	-0.003
1.0	Delta_i = .25	2	0.027	0.016	0.002
1.0	Delta_i = .25	4	0.057	0.036	0.008
1.0	Delta_i = .25	6	0.073	0.049	-0.003
1.0	Delta_i = .25	8	0.103	0.087	0.004
1.0	Delta_i = .25	10	0.122	0.121	-0.005
1.0	Delta_i = .25	12	0.151	0.165	-0.001
1.0	Delta_i = .25	14	0.182	0.200	0.010
1.0	Delta_i = .25	16	0.198	0.212	0.003
1.0	Delta_i = .25	18	0.226	0.237	0.001
1.0	Delta_i = .25	20	0.248	0.245	NA
1.0	Delta_i = .50	0	-0.002	-0.001	-0.002
1.0	Delta_i = .50	2	0.056	0.009	0.006
1.0	Delta_i = .50	4	0.100	0.008	0.001
1.0	Delta_i = .50	6	0.149	0.015	0.000
1.0	Delta_i = .50	8	0.206	0.060	0.006
1.0	Delta_i = .50	10	0.254	0.242	0.003
1.0	Delta_i = .50	12	0.301	0.436	0.002
1.0	Delta_i = .50	14	0.356	0.490	0.007
1.0	Delta_i = .50	16	0.402	0.488	0.003
1.0	Delta_i = .50	18	0.454	0.496	0.000
1.0	Delta_i = .50	20	0.499	0.496	NA
1.5	Delta_i = .25	0	0.000	0.000	0.000
1.5	Delta_i = .25	2	0.025	0.017	0.000
1.5	Delta_i = .25	4	0.049	0.037	0.001
1.5	Delta_i = .25	6	0.070	0.061	-0.005
1.5	Delta_i = .25	8	0.100	0.105	0.000
1.5	Delta_i = .25	10	0.120	0.144	-0.006
1.5	Delta_i = .25	12	0.151	0.192	-0.004
1.5	Delta_i = .25	14	0.176	0.212	0.003
1.5	Delta_i = .25	16	0.203	0.228	0.003
1.5	Delta_i = .25	18	0.228	0.242	0.001
1.5	Delta_i = .25	20	0.252	0.253	NA

slope_multiplier	target_Delta_i	n_dif	Naive	RDIF	True
1.5	Delta_i = .50	0	0.004	0.004	0.004
1.5	Delta_i = .50	2	0.047	-0.002	-0.003
1.5	Delta_i = .50	4	0.100	0.004	-0.001
1.5	Delta_i = .50	6	0.148	0.007	-0.003
1.5	Delta_i = .50	8	0.197	0.055	-0.003
1.5	Delta_i = .50	10	0.254	0.302	0.005
1.5	Delta_i = .50	12	0.300	0.470	0.000
1.5	Delta_i = .50	14	0.344	0.483	-0.009
1.5	Delta_i = .50	16	0.402	0.497	0.001
1.5	Delta_i = .50	18	0.447	0.493	-0.002
1.5	Delta_i = .50	20	0.498	0.497	NA

DTF Effect Bias

```

delta_bias <- sim_delta |>
  mutate(Bias = delta_hat - true_Delta) |>
  summarize(
    bias = mean_or_na(Bias),
    true_Delta = mean_or_na(true_Delta),
    .by = c(
      method, slope_multiplier, slope_cell, slope_cell_public,
      target_Delta_i, target_Delta_i_num, target_Delta_i_public, n_dif, p_dif))

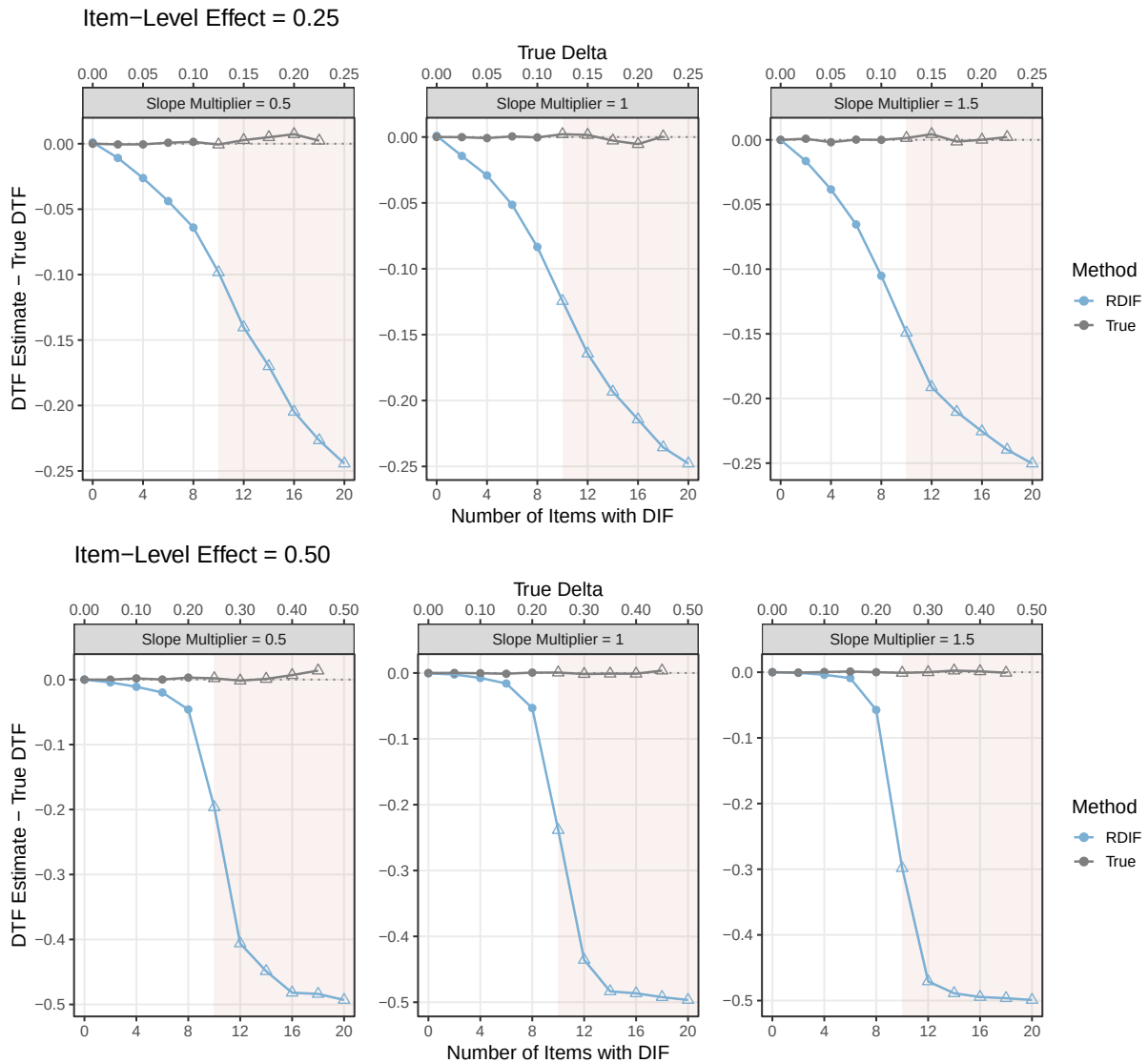
delta_bias_plots <- plot_by_effect(delta_bias, function(plot_data) {
  ggplot(plot_data, aes(x = n_dif, y = bias, color = method, group = method,
    shape = p_dif >= 0.5)) +
    geom_rect(
      data = breakdown_band(plot_data),
      aes(xmin = xmin, xmax = xmax, ymin = ymin, ymax = ymax),
      inherit.aes = FALSE,
      fill = "#B85C38",
      alpha = 0.08) +
    geom_hline(yintercept = 0, linetype = 3, color = "grey50") +
    geom_point(size = 2.4, na.rm = TRUE) +
    geom_line(linewidth = 0.8, na.rm = TRUE) +
    scale_color_manual(values = method_cols, drop = FALSE) +
    scale_shape_manual(
      values = c("FALSE" = 16, "TRUE" = 2),
      guide = "none") +

```

```

ndif_x_scale(plot_data) +
coord_cartesian(clip = "off") +
sim2_public_facets(free_y = TRUE) +
labs(
  x = "Number of Items with DIF",
  y = "DTF Estimate - True DTF",
  color = "Method") +
public_plot_theme
})

```



```

delta_bias_wide <- delta_bias |>
  mutate(method = as.character(method)) |>
  arrange(slope_multiplier, target_Delta_i, n_dif, method) |>
  select(method, slope_multiplier, target_Delta_i, n_dif, bias) |>
  pivot_wider(names_from = method, values_from = bias)

knitr::kable(delta_bias_wide, digits = 3)

```

slope_multiplier	target_Delta_i	n_dif	RDIF	True
0.5	Delta_i = .25	0	0.001	0.000
0.5	Delta_i = .25	2	-0.011	-0.001
0.5	Delta_i = .25	4	-0.026	0.000
0.5	Delta_i = .25	6	-0.044	0.001
0.5	Delta_i = .25	8	-0.064	0.001
0.5	Delta_i = .25	10	-0.098	-0.001
0.5	Delta_i = .25	12	-0.140	0.003
0.5	Delta_i = .25	14	-0.170	0.005
0.5	Delta_i = .25	16	-0.205	0.007
0.5	Delta_i = .25	18	-0.227	0.002
0.5	Delta_i = .25	20	-0.244	NA
0.5	Delta_i = .50	0	0.000	0.000
0.5	Delta_i = .50	2	-0.004	0.000
0.5	Delta_i = .50	4	-0.011	0.002
0.5	Delta_i = .50	6	-0.020	0.000
0.5	Delta_i = .50	8	-0.046	0.003
0.5	Delta_i = .50	10	-0.197	0.002
0.5	Delta_i = .50	12	-0.406	-0.002
0.5	Delta_i = .50	14	-0.449	0.001
0.5	Delta_i = .50	16	-0.482	0.007
0.5	Delta_i = .50	18	-0.484	0.014
0.5	Delta_i = .50	20	-0.493	NA
1.0	Delta_i = .25	0	0.001	0.000
1.0	Delta_i = .25	2	-0.014	0.000
1.0	Delta_i = .25	4	-0.029	-0.001
1.0	Delta_i = .25	6	-0.051	0.000
1.0	Delta_i = .25	8	-0.083	0.000
1.0	Delta_i = .25	10	-0.124	0.002
1.0	Delta_i = .25	12	-0.164	0.002
1.0	Delta_i = .25	14	-0.193	-0.003
1.0	Delta_i = .25	16	-0.214	-0.005
1.0	Delta_i = .25	18	-0.236	0.000

slope_multiplier	target_Delta_i	n_dif	RDIF	True
1.0	Delta_i = .25	20	-0.248	NA
1.0	Delta_i = .50	0	-0.001	0.000
1.0	Delta_i = .50	2	-0.002	0.000
1.0	Delta_i = .50	4	-0.008	0.000
1.0	Delta_i = .50	6	-0.016	-0.001
1.0	Delta_i = .50	8	-0.053	0.001
1.0	Delta_i = .50	10	-0.239	0.001
1.0	Delta_i = .50	12	-0.436	-0.002
1.0	Delta_i = .50	14	-0.484	-0.001
1.0	Delta_i = .50	16	-0.486	-0.001
1.0	Delta_i = .50	18	-0.492	0.004
1.0	Delta_i = .50	20	-0.497	NA
1.5	Delta_i = .25	0	0.000	0.000
1.5	Delta_i = .25	2	-0.016	0.001
1.5	Delta_i = .25	4	-0.038	-0.002
1.5	Delta_i = .25	6	-0.065	0.000
1.5	Delta_i = .25	8	-0.105	0.000
1.5	Delta_i = .25	10	-0.149	0.001
1.5	Delta_i = .25	12	-0.191	0.004
1.5	Delta_i = .25	14	-0.210	-0.001
1.5	Delta_i = .25	16	-0.225	0.000
1.5	Delta_i = .25	18	-0.240	0.002
1.5	Delta_i = .25	20	-0.250	NA
1.5	Delta_i = .50	0	0.000	0.000
1.5	Delta_i = .50	2	-0.001	-0.001
1.5	Delta_i = .50	4	-0.004	0.000
1.5	Delta_i = .50	6	-0.009	0.001
1.5	Delta_i = .50	8	-0.057	0.000
1.5	Delta_i = .50	10	-0.298	-0.001
1.5	Delta_i = .50	12	-0.471	0.000
1.5	Delta_i = .50	14	-0.489	0.002
1.5	Delta_i = .50	16	-0.495	0.001
1.5	Delta_i = .50	18	-0.496	-0.001
1.5	Delta_i = .50	20	-0.499	NA

Rejection Rates

```

sim_power <- sim_delta |>
  summarize(
    rejection_rate = mean_or_na(as.numeric(rejected)),
    true_Delta = mean_or_na(true_Delta),
    .by = c(
      method, slope_multiplier, slope_cell, slope_cell_public,
      target_Delta_i, target_Delta_i_num, target_Delta_i_public,
      n_dif, p_dif, n_items)) |>
  mutate(
    rejection_rate = ifelse(
      method == "True" & n_dif == n_items,
      NA_real_,
      rejection_rate))

```

```

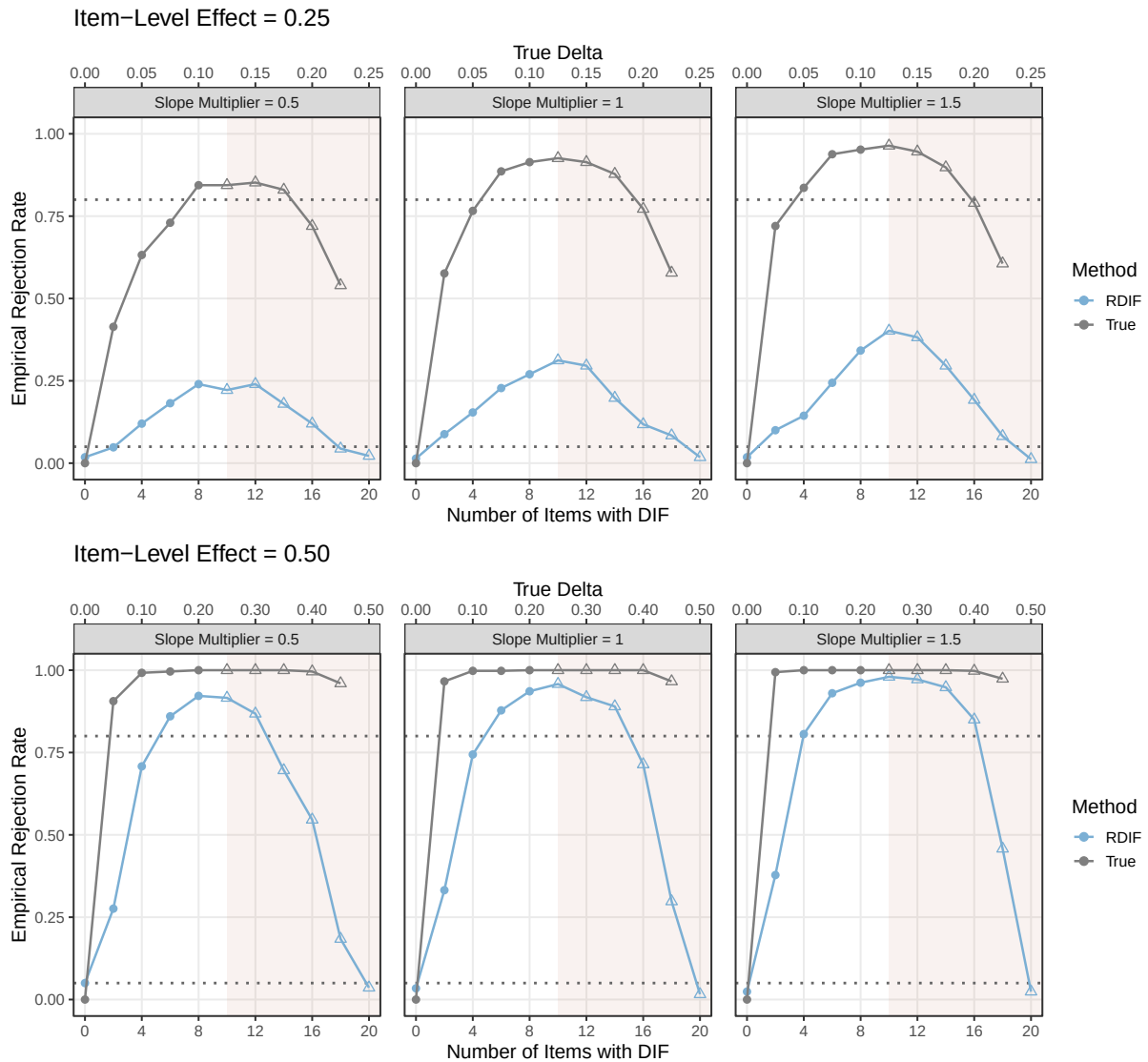
rejection_plots <- plot_by_effect(
  sim_power,
  function(plot_data) {
    ggplot(
      plot_data,
      aes(x = n_dif, y = rejection_rate, color = method, group = method,
          shape = p_dif >= 0.5)) +
    geom_rect(
      data = breakdown_band(plot_data),
      aes(xmin = xmin, xmax = xmax, ymin = ymin, ymax = ymax),
      inherit.aes = FALSE,
      fill = "#B85C38",
      alpha = 0.08) +
    geom_hline(yintercept = alpha, linetype = 3, linewidth = 0.9, color = "grey40") +
    geom_hline(yintercept = 0.80, linetype = 3, linewidth = 0.9, color = "grey40") +
    geom_point(size = 2.4, na.rm = TRUE) +
    geom_line(linewidth = 0.8, na.rm = TRUE) +
    scale_y_continuous(limits = c(0, 1), breaks = seq(0, 1, .25)) +
    scale_color_manual(values = method_cols, drop = FALSE) +
    scale_shape_manual(
      values = c("FALSE" = 16, "TRUE" = 2),
      guide = "none") +
    ndif_x_scale(plot_data) +
    coord_cartesian(clip = "off") +
    sim2_public_facets() +
    labs(
      x = "Number of Items with DIF",

```

```

    y = "Empirical Rejection Rate",
    color = "Method") +
    public_plot_theme
  },
  file_prefix = "rejection_rates_by_true_delta")

```



```

sim_power_wide <- sim_power |>
  select(method, slope_multiplier, target_Delta_i, n_dif, true_Delta,
    rejection_rate) |>
  pivot_wider(names_from = method, values_from = rejection_rate)

```

```
knitr::kable(sim_power_wide, digits = 3)
```

slope_multiplier	target_Delta_i	n_dif	true_Delta	RDIF	True
0.5	Delta_i = .25	0	0.000	0.018	0.000
0.5	Delta_i = .25	2	0.025	0.048	0.414
0.5	Delta_i = .25	4	0.050	0.120	0.632
0.5	Delta_i = .25	6	0.075	0.182	0.730
0.5	Delta_i = .25	8	0.100	0.240	0.844
0.5	Delta_i = .25	10	0.125	0.222	0.844
0.5	Delta_i = .25	12	0.150	0.240	0.852
0.5	Delta_i = .25	14	0.175	0.180	0.830
0.5	Delta_i = .25	16	0.200	0.120	0.720
0.5	Delta_i = .25	18	0.225	0.044	0.540
0.5	Delta_i = .25	20	0.250	0.022	NA
0.5	Delta_i = .50	0	0.000	0.050	0.000
0.5	Delta_i = .50	2	0.050	0.276	0.906
0.5	Delta_i = .50	4	0.100	0.708	0.992
0.5	Delta_i = .50	6	0.150	0.860	0.996
0.5	Delta_i = .50	8	0.200	0.922	1.000
0.5	Delta_i = .50	10	0.250	0.916	1.000
0.5	Delta_i = .50	12	0.300	0.868	1.000
0.5	Delta_i = .50	14	0.350	0.696	1.000
0.5	Delta_i = .50	16	0.400	0.546	0.996
0.5	Delta_i = .50	18	0.450	0.184	0.960
0.5	Delta_i = .50	20	0.500	0.036	NA
1.0	Delta_i = .25	0	0.000	0.014	0.000
1.0	Delta_i = .25	2	0.025	0.088	0.576
1.0	Delta_i = .25	4	0.050	0.154	0.766
1.0	Delta_i = .25	6	0.075	0.228	0.886
1.0	Delta_i = .25	8	0.100	0.270	0.914
1.0	Delta_i = .25	10	0.125	0.312	0.926
1.0	Delta_i = .25	12	0.150	0.296	0.914
1.0	Delta_i = .25	14	0.175	0.198	0.878
1.0	Delta_i = .25	16	0.200	0.118	0.772
1.0	Delta_i = .25	18	0.225	0.084	0.578
1.0	Delta_i = .25	20	0.250	0.018	NA
1.0	Delta_i = .50	0	0.000	0.034	0.000
1.0	Delta_i = .50	2	0.050	0.332	0.966
1.0	Delta_i = .50	4	0.100	0.744	0.998
1.0	Delta_i = .50	6	0.150	0.878	0.998

slope_multiplier	target_Delta_i	n_dif	true_Delta	RDIF	True
1.0	Delta_i = .50	8	0.200	0.936	1.000
1.0	Delta_i = .50	10	0.250	0.958	1.000
1.0	Delta_i = .50	12	0.300	0.918	1.000
1.0	Delta_i = .50	14	0.350	0.890	1.000
1.0	Delta_i = .50	16	0.400	0.714	1.000
1.0	Delta_i = .50	18	0.450	0.298	0.966
1.0	Delta_i = .50	20	0.500	0.016	NA
1.5	Delta_i = .25	0	0.000	0.018	0.000
1.5	Delta_i = .25	2	0.025	0.100	0.720
1.5	Delta_i = .25	4	0.050	0.144	0.836
1.5	Delta_i = .25	6	0.075	0.244	0.938
1.5	Delta_i = .25	8	0.100	0.342	0.952
1.5	Delta_i = .25	10	0.125	0.402	0.964
1.5	Delta_i = .25	12	0.150	0.382	0.946
1.5	Delta_i = .25	14	0.175	0.296	0.898
1.5	Delta_i = .25	16	0.200	0.192	0.790
1.5	Delta_i = .25	18	0.225	0.082	0.606
1.5	Delta_i = .25	20	0.250	0.012	NA
1.5	Delta_i = .50	0	0.000	0.024	0.000
1.5	Delta_i = .50	2	0.050	0.378	0.994
1.5	Delta_i = .50	4	0.100	0.806	1.000
1.5	Delta_i = .50	6	0.150	0.930	1.000
1.5	Delta_i = .50	8	0.200	0.962	1.000
1.5	Delta_i = .50	10	0.250	0.980	1.000
1.5	Delta_i = .50	12	0.300	0.972	1.000
1.5	Delta_i = .50	14	0.350	0.948	1.000
1.5	Delta_i = .50	16	0.400	0.850	0.998
1.5	Delta_i = .50	18	0.450	0.458	0.974
1.5	Delta_i = .50	20	0.500	0.024	NA

Coverage

Coverage is based on the stored indicator from `data_sim2_target_delta()`, which checks whether the nominal 95% Wald interval for `delta_hat` contains the true DTF value, `true_Delta`.

```
sim_coverage <- sim_delta |>
  summarize(
    coverage = mean_or_na(as.numeric(covered)),
```

```

mean_width = mean_or_na(2 * qnorm(.975) * delta_se),
mean_se = mean_or_na(delta_se),
true_Delta = mean_or_na(true_Delta),
.by = c(
  method, slope_multiplier, slope_cell, slope_cell_public,
  target_Delta_i, target_Delta_i_num, target_Delta_i_public, n_dif, p_dif))

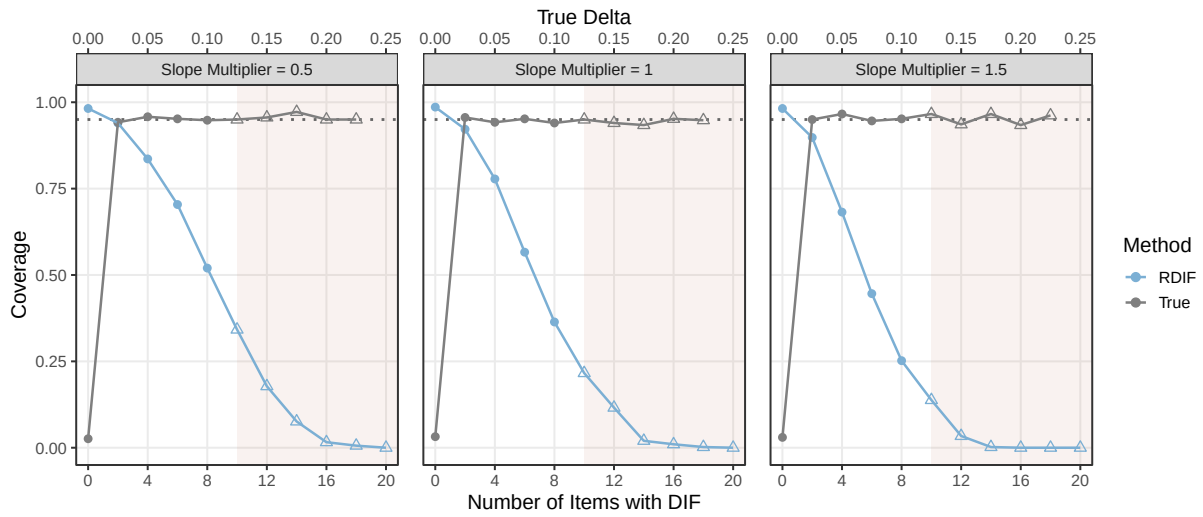
```

```

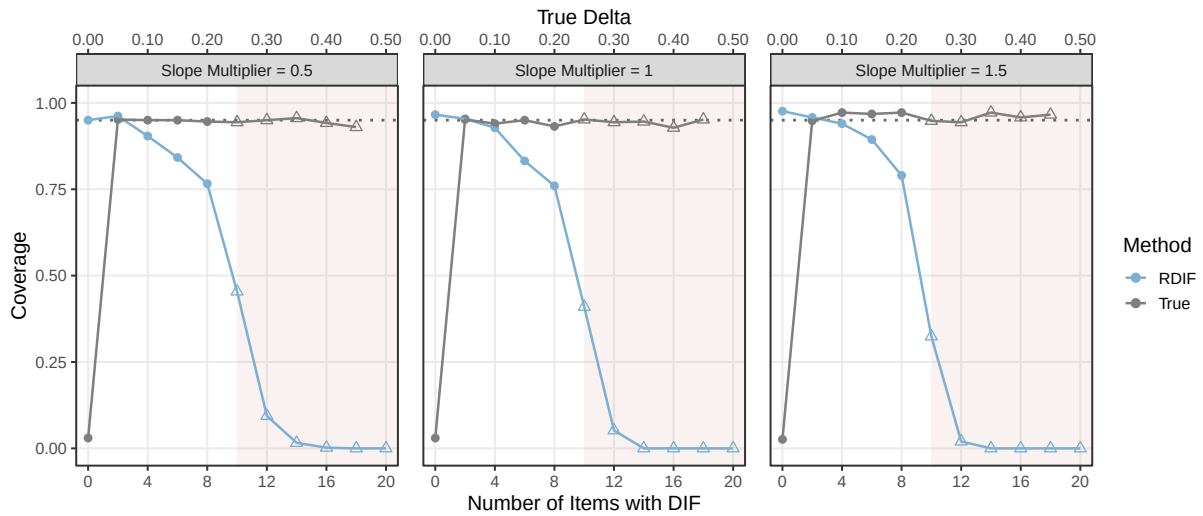
coverage_plots <- plot_by_effect(
  sim_coverage,
  function(plot_data) {
    ggplot(
      plot_data,
      aes(x = n_dif, y = coverage, color = method, group = method,
          shape = p_dif >= 0.5)) +
    geom_rect(
      data = breakdown_band(plot_data),
      aes(xmin = xmin, xmax = xmax, ymin = ymin, ymax = ymax),
      inherit.aes = FALSE,
      fill = "#B85C38",
      alpha = 0.08) +
    geom_hline(yintercept = 0.95, linetype = 3, linewidth = 0.9, color = "grey40") +
    geom_point(size = 2.4, na.rm = TRUE) +
    geom_line(linewidth = 0.8, na.rm = TRUE) +
    scale_y_continuous(limits = c(0, 1), breaks = seq(0, 1, .25)) +
    scale_color_manual(values = method_cols, drop = FALSE) +
    scale_shape_manual(
      values = c("FALSE" = 16, "TRUE" = 2),
      guide = "none") +
    ndif_x_scale(plot_data) +
    coord_cartesian(clip = "off") +
    sim2_public_facets() +
    labs(
      x = "Number of Items with DIF",
      y = "Coverage",
      color = "Method") +
    public_plot_theme
  },
  file_prefix = "coverage_by_true_delta")

```

Item-Level Effect = 0.25



Item-Level Effect = 0.50



```
sim_coverage_wide <- sim_coverage |>
  select(method, slope_multiplier, target_Delta_i, n_dif, coverage) |>
  pivot_wider(names_from = method, values_from = coverage)

knitr::kable(sim_coverage_wide, digits = 3)
```

slope_multiplier	target_Delta_i	n_dif	RDIF	True
0.5	Delta_i = .25	0	0.982	0.026
0.5	Delta_i = .25	2	0.940	0.942

slope_multiplier	target_Delta_i	n_dif	RDIF	True
0.5	Delta_i = .25	4	0.836	0.958
0.5	Delta_i = .25	6	0.704	0.952
0.5	Delta_i = .25	8	0.520	0.948
0.5	Delta_i = .25	10	0.342	0.950
0.5	Delta_i = .25	12	0.178	0.956
0.5	Delta_i = .25	14	0.076	0.972
0.5	Delta_i = .25	16	0.016	0.950
0.5	Delta_i = .25	18	0.006	0.950
0.5	Delta_i = .25	20	0.000	NA
0.5	Delta_i = .50	0	0.950	0.030
0.5	Delta_i = .50	2	0.962	0.952
0.5	Delta_i = .50	4	0.904	0.950
0.5	Delta_i = .50	6	0.842	0.950
0.5	Delta_i = .50	8	0.766	0.946
0.5	Delta_i = .50	10	0.454	0.944
0.5	Delta_i = .50	12	0.094	0.950
0.5	Delta_i = .50	14	0.016	0.956
0.5	Delta_i = .50	16	0.002	0.942
0.5	Delta_i = .50	18	0.000	0.930
0.5	Delta_i = .50	20	0.000	NA
1.0	Delta_i = .25	0	0.986	0.032
1.0	Delta_i = .25	2	0.922	0.956
1.0	Delta_i = .25	4	0.778	0.942
1.0	Delta_i = .25	6	0.566	0.952
1.0	Delta_i = .25	8	0.364	0.940
1.0	Delta_i = .25	10	0.216	0.950
1.0	Delta_i = .25	12	0.116	0.940
1.0	Delta_i = .25	14	0.020	0.934
1.0	Delta_i = .25	16	0.010	0.952
1.0	Delta_i = .25	18	0.002	0.948
1.0	Delta_i = .25	20	0.000	NA
1.0	Delta_i = .50	0	0.966	0.030
1.0	Delta_i = .50	2	0.954	0.952
1.0	Delta_i = .50	4	0.928	0.940
1.0	Delta_i = .50	6	0.832	0.950
1.0	Delta_i = .50	8	0.760	0.932
1.0	Delta_i = .50	10	0.410	0.952
1.0	Delta_i = .50	12	0.052	0.944
1.0	Delta_i = .50	14	0.000	0.946
1.0	Delta_i = .50	16	0.000	0.928
1.0	Delta_i = .50	18	0.000	0.952

slope_multiplier	target_Delta_i	n_dif	RDIF	True
1.0	Delta_i = .50	20	0.000	NA
1.5	Delta_i = .25	0	0.982	0.030
1.5	Delta_i = .25	2	0.898	0.950
1.5	Delta_i = .25	4	0.682	0.966
1.5	Delta_i = .25	6	0.446	0.946
1.5	Delta_i = .25	8	0.252	0.952
1.5	Delta_i = .25	10	0.138	0.966
1.5	Delta_i = .25	12	0.034	0.936
1.5	Delta_i = .25	14	0.002	0.966
1.5	Delta_i = .25	16	0.000	0.934
1.5	Delta_i = .25	18	0.000	0.962
1.5	Delta_i = .25	20	0.000	NA
1.5	Delta_i = .50	0	0.976	0.026
1.5	Delta_i = .50	2	0.958	0.948
1.5	Delta_i = .50	4	0.940	0.972
1.5	Delta_i = .50	6	0.894	0.968
1.5	Delta_i = .50	8	0.790	0.972
1.5	Delta_i = .50	10	0.324	0.948
1.5	Delta_i = .50	12	0.020	0.944
1.5	Delta_i = .50	14	0.000	0.972
1.5	Delta_i = .50	16	0.000	0.958
1.5	Delta_i = .50	18	0.000	0.966
1.5	Delta_i = .50	20	0.000	NA